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Date: 06/30 – 07/03

Weekly Report

Week: 06/30 – 07/03

Subject of This Week:

- **Bookstore Project – Database Normalization & Django Web Interface**
 - Refactored Bookstore database to Third Normal Form (3NF):
 - Created authors table with first_name, last_name, bio, birth_year.
 - Created book_authors association table for many-to-many relationship.
 - Added publisher and quantity_on_hand to books table.
 - Split orders into orders header and order_details tables.
 - Changed order_id to numeric primary key with order_number as business key.
 - Added NOT NULL constraints on all non-key columns.
 - Built Django web interface for Bookstore Project:
 - Created search page supporting title, author (first/last name), and genre search.
 - Created order page with form for name, email, quantity, and mobile.
 - Integrated stock management (auto-deduct on order).
 - Admin panel for managing books, authors, and orders.
 - Imported 50 books from Open Library API using custom management command.
 - Updated dbdiagram.io ERD to match new database schema.
 - Deployed development server and tested end-to-end flow.
 - Added README with setup instructions.
- **MIT 6.0001 Python Lectures**
 - Completed Lectures 1 through 16.

Notes:

- Database normalization eliminates data redundancy and allows multiple authors per book.
- Django search uses Q objects for multi-field searching.
- Orders are stored with header/details pattern for scalability.
- Stock updates automatically when orders are placed.
- Email sending causes a few seconds delay (needs async improvement).

Plan for Next Week:

- Deploy Django app to dev server.
- Add user authentication (login/register).
- Improve UI with better styling.

- Add order history page.

Hours Worked: 30

Intern Checklist:

1. Receive the internship offer letter
2. Accept the terms of the offer letter
 - 2.1. Create a Telegram account and be added into group chat
 - 2.2. Create a LinkedIn account (Connect with Verbus, and other interns in the program. Then join MMIS Professionals and Mostly Medicaid (Medicaid Professionals))
 - 2.3. Create a GitHub account
 - 2.4. Please use Mermaid in your documentation.
3. Recommended AI assistant is CLAUDE
4. Attend the daily standup calls
5. Take notes and produce weekly report (due Friday's, Saturdays at the latest)
 - 5.1. PDF (EMRTS_Weekly_report_date_firstname_lastname.pdf)
 - 5.2. JSON (EMRTS_Weekly_report_date_firstname_lastname.json)
6. Watch the Python Lectures from MIT (Lectures 1-16)
 - 6.1. <https://ocw.mit.edu/courses/6-0001-introduction-to-computer-science-and-programming-in-python-fall-2016/>
10. Work on the Assigned Project
 - 10.1. State Capitals Project - completed
 - 10.2. 4-Gallon Water Bucket Problem - completed
 - 10.3. TABIND Scrabble Project - completed
 - 10.4. FizzBuzz Project - completed
 - 10.5. Provider Lookup Project - completed
 - 10.6. Bookstore Project - in progress

EMRTS

<https://emrts.us/>

GitHub

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